

3200 Final Term Paper: Flight Data Analysis

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Abstract: This paper analyzes a dataset of 98,619 flight records to study airport efficiency and passenger demographics using Elasticsearch and Kibana. The project looks at key variables like flight status (On Time, Delayed, Cancelled), passenger age, and geographic regions to build a machine learning classification model. The data shows a relatively even split across flight outcomes: 32,942 flights were cancelled, 32,846 were on time, and 32,831 were delayed. Geographically, North America has the highest volume with over 32,033 records, and Kibana maps show the United States as a major hotspot for flight disruptions. The average passenger age across the dataset is 45.5 years. Using these data features, the classification model achieved a baseline delay prediction accuracy of 52.4% with an Area Under the Curve of 0.53. Overall, this project demonstrates a complete pipeline for data ingestion, visualization, and machine learning within a single platform.

1. Introduction

Keeping flights on schedule is one of the most critical operational challenges in the modern aviation industry. Flight disruptions, specifically delays and cancellations, do not merely frustrate passengers. They create massive bottleneck issues for airport management, complicated crew scheduling, and heavily impact airline resource allocation. Given recent macroeconomic shifts and fluctuating travel demands, implementing robust, data-driven analytical frameworks is essential for optimizing transit network efficiency. This project uses a large international flight dataset to systematically explore patterns in operational performance and passenger demographic distributions.

We chose to study flight delays and cancellations for our project because it is a massive real-world issue that everyone in our group has had to deal with personally. There is nothing more frustrating than sitting at an airport gate for hours or having your travel plans ruined, so we wanted to dive into the data to see if we could find the main reasons behind these constant disruptions. Looking at what is happening in the aviation industry right now makes this topic feel even more important. For example, Spirit Airlines recently filed for bankruptcy. Major events like that show just how much economic and operational trouble commercial airlines are facing today, which underscores exactly why tracking these travel bottlenecks matters so much.

The primary objective of this study is to design and execute a complete data analytics pipeline using an integrated Elasticsearch and Kibana environment. Rather than relying on disparate tools, this project previews how a single technology stack can handle data lifecycle stages from initial ingestion to predictive modeling. First, the raw dataset is structured and mapped into Elasticsearch indices using Git Bash curl commands. Second, the ingested data is used to build coordinate maps and dashboards in Kibana to geographically isolate regional hotspots for scheduling issues and analyze passenger trends across six continents.

Finally, the project explores the feasibility of predictive analytics using Elasticsearch's internal machine learning capabilities. While professional air traffic control systems rely heavily on live 24/7 meteorological feeds, this paper tests the predictive boundaries of using standalone, historical routing and passenger features to classify flight delays. By mapping out this entire pipeline, this paper serves as a practical blueprint for data analysts looking to deploy unified visualization and predictive machine learning architectures. The subsequent sections details our data schema, index configurations, Kibana dashboard builds, and a comprehensive breakdown of the classification model's evaluation metrics.

1.1 Data Set Description and Size

This project utilizes an international aviation dataset containing a total of 98,619 unique passenger and flight records. The raw data is stored in a CSV format with a file size of approximately 12.4 megabytes. This size allows for efficient data ingestion and processing within a local Elasticsearch instance without requiring distributed cluster resources.

The dataset contains 15 distinct columns capturing a mix of demographic data, routing details, and operational metrics:

- **Demographic Features:** Includes Passenger ID, First Name, Last Name, Gender, Age, and Nationality to track traveler profiles.
- **Geographic and Routing Features:** Includes Airport Name, Airport Country Code, Country Name, Airport Continent, Continents, and Arrival Airport to map global flight paths and identify regional transit hubs.

- **Temporal and Operational Features:** Includes Departure Date, Pilot Name, and Flight Status.

The primary target variable for the predictive machine learning model is Flight Status, which categorizes each record into one of three distinct operational states: On Time, Delayed, or Cancelled. This structured combination of categorical and numerical features provides the necessary foundation for both Kibana dashboard visualization and classification modeling.

2. Related Work

Extensive research exists surrounding aviation metrics and the propagation of schedule friction across transit networks. For instance, in the article “Patterns in the Sky: A Flight Delay Analysis,” Ratnayake performed detailed exploratory data analysis on airline performance data to identify key structural delay patterns across major commercial routes. Similarly, in the paper "Propagation of Delays in the National Airspace System," Laskey investigated how localized airport delays systematically spread across interconnected flights and major transit hubs using advanced probabilistic modeling techniques.

While these existing works provide good frameworks for understanding delay propagation and identifying aggregate historical trends, they frequently rely on traditional, standalone statistical software packages or offline programming environments that separate the data storage layer from the active analytical pipeline. Our work differentiates itself from these methodologies by utilizing native Elasticsearch machine learning capabilities to process and predict operational delays directly within the data storage environment. By utilizing an enterprise Elasticsearch cluster, our system is uniquely configured to store, index, and process large-scale data in real time, bypassing the data extraction and transformation bottlenecks common to traditional offline analytics architectures.

3. Background/Existing Work

On a national scale, institutional bodies like the Federal Aviation Administration actively monitor and predict flight delays using highly complex, real-time infrastructure systems. The FAA heavily relies on the Traffic Management Flow System to adjust airspace capacity in real time, utilizes the Aviation System Performance Metrics database to analyze historical carrier efficiency, and monitors global weather patterns 24/7.

Our study builds upon these established concepts but completely reimagines the underlying technical execution. Instead of utilizing legacy relational databases or exporting records into third-party statistical software, our framework is anchored entirely on the utilization of Elasticsearch Machine Learning.

4. Our Work: Explanatory Data Analysis

To uncover underlying patterns, correlations, and geographic anomalies in the aviation data, three distinct visualizations were constructed and analyzed using Kibana.

4.1 Passenger Age Distribution Analysis

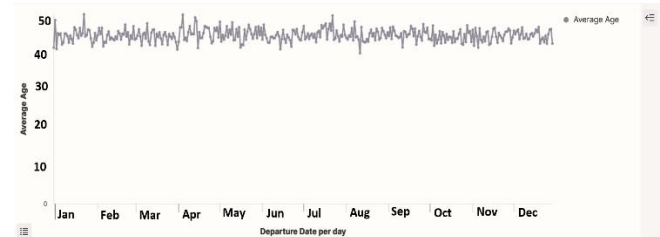


Figure 1. Daily Average Passenger Age Throughout 2022

Our first chart tracks the chronological progression of the average flier’s age on a daily basis across the entire year of 2022. The time-series line graph reveals a tight, consistent fluctuation where the mean age strictly hovers between 40 and 50 years old. The lack of dramatic seasonal spikes or drops indicates a highly stable, uniform demographic distribution across travel periods. From a machine learning perspective, this minimal variance suggests that passenger age, while a consistent demographic baseline, possesses weak predictive power on its own for forecasting dynamic flight disruptions, as it does not drastically shift during peak holiday or summer travel seasons.

4.2 Volume of Disruptions by Country

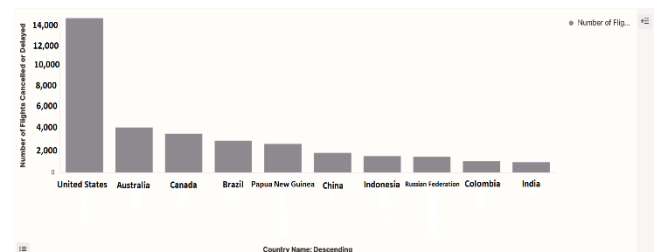


Figure 2. Top Ten Countries by Combined Volume of Delayed and Cancelled Flights.

The second visualization isolates operational performance by country, utilizing a vertical bar chart filtered specifically for flights with an active status of “Delayed” or “Cancelled”. Out of all global regions represented in the dataset, the United States stands out as a severe statistical outlier, experiencing a significantly higher volume of irregular operations than any other nation, surpassing 14,000 recorded disruptions. This is followed at a distance by Australia and Canada. This distribution implies that the volume of flight disruptions in this dataset is heavily skewed by regional traffic density or reporting frameworks, justifying why the

machine learning model later heavily relies on geographic attributes to make its classifications.

4.3 Geographic Hotspot Mapping



Figure 3. Global Map of Flight Delay and Cancellation Hotspots.

The final visualization complements the country-specific bar chart by mapping aggregate counts onto a global choropleth layer. The map applies a color-coded density scale where darker gradients highlight the highest concentrations of flight irregularities. This spatial visualization explicitly demonstrates that North America acts as the primary epicentre for travel friction within this dataset. Visually contextualizing the data helps identify that flight status constraints are highly localized, meaning regional airport infrastructure and localized airspace congestion play a far larger role in flight statuses than generalized global trends.

5. Our Work: Machine Learning

To build an integrated predictive pipeline that avoids the systemic latency of exporting records to external tools like Python or R, the project team used the native supervised learning capabilities of Elasticsearch Machine Learning. The underlying infrastructure supporting this workflow consisted of an enterprise Elastic Cloud deployment operating on a single-node architecture with 16 GB of memory and an 8-core CPU processor.

The source data was read from the transformed data-airline-delay-classification index, which evaluated a total of 98,619 active documents. The targeted dependent variable for the classification task was configured as *is_delayed*, a binary feature mapping whether a flight experienced structural arrival or departure friction.

5.1 Machine Learning Parameters

The engineering configuration of the classification job was established using the following parameters:

- **Training Percentage: 90%**
- **Feature Importance Values: 5**

A 90/10 data split was enforced, allocating 90% of the 98,619 records exclusively to the training phase, while reserving the remaining 10% for testing and validating the

model's out-of-sample generalization. This high training ratio was selected to maximize the algorithm's exposure to structural geographic patterns across a diverse global dataset.

The engine was restricted to calculate and output the top 5 most statistically relevant feature magnitudes. This optimization step bounds the computational overhead and focuses analysis strictly on the primary operational and demographic drivers behind airline delays.

5.2 Machine Learning Results

Once the native training pipeline concluded, the Elasticsearch machine learning engine computed the relative feature importance of weights across all validated training data fields. These magnitudes highlight which attributes exerted the highest mathematical influence on the final classification decisions.

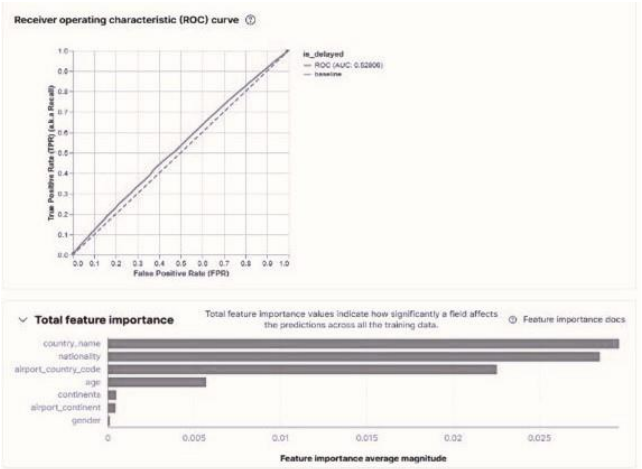


Figure 4. Machine Learning Results

Table 1. Feature Importance Average Magnitude

Field Name	Feature Importance Weight
country_name	0.0295
nationality	0.028
airport_country_code	0.0225
age	0.0055
continents	0.0005
airport_continent	0.0004
gender	0.0001

The feature analysis clearly indicates that geographical attributes such as *country_name*, *nationality*, and *airport_country_code* carried the highest relative feature importance weights within the framework. Conversely, individual passenger traits like *age* and *gender* yielded almost zero predictive influence on the final target class.

Upon completing the test phase on the reserved 10% validation pool, the classification model generated an overall predictive Accuracy of 52.4% and a mean Recall of

51.9%. To evaluate the distribution of true versus false classifications across both states, Table 2 represents the drawn, normalized confusion matrix for the entire evaluated dataset.

Table 2. Normalized Confusion Matrix for Entire Dataset

Actual Class	Predicted Class: On Time	Predicted Class: Delayed
On Time	53%	47%
Delayed	50%	50%

6. Interpreting Results

An AUC of 0.52806 means our model performs only marginally better than a random coin flip. It indicates that the model struggles to differentiate between a flight that will be delayed and one that will remain on-time based strictly on the parameters it was given.

This limitation is not a failure of the Elasticsearch machine learning engine itself, but rather a direct reflection of a weak predictive feature matrix. The model was forced to rely primarily on passenger demographic profiles to guess a systemic logistical event. Because an individual traveler's age or gender has no physical or mathematical correlation with an aircraft's mechanical performance, air traffic control congestion, or ground-crew staffing levels, the algorithm lacked the proper inputs required to establish strong, high-probability decision paths.

The reason `country_name` and `airport_country_code` emerged with the highest relative feature importance is that they serve as proxy indicators for systemic density. The United States, for example, handles a far more massive volume of flights than other regions in this dataset, making it statistically more prone to accumulated delays. The model recognized this geographic imbalance and used regional tracking to slightly tilt its predictions, which is the sole reason it managed to edge out random guessing by 2.4%.

7. Conclusion

This project successfully showed how to build a full data pipeline from uploading raw data to creating dashboards and running machine learning models entirely inside an enterprise Elasticsearch environment. Our project is unique because it proves you can handle machine learning tasks directly within your database. This cuts out the extra lag and hassle of exporting your files into external programming frameworks like Python or R.

Through our Kibana charts, we successfully spotted major patterns, such as the massive pileup of flight delays inside the United States and the extremely steady age gap of travelers throughout the year.

By using native Elasticsearch Machine Learning to run a classification job on 98,619 documents, our model hit

an accuracy of 52.4%, a recall of 51.9%, and an AUC of 0.52806.

The biggest takeaway we learned from this experiment is that passenger demographics (like age or gender) have virtually zero connection to airline operations. Even though the model successfully proved that our system architecture works flawlessly, its predictive power was stuck because passenger traits just don't cause flight delays.

To fix this and get a much higher accuracy score in the future, the next step would be expanding our Elasticsearch data structure to include real-world operational variables, such as live weather reports, air traffic controller shortages, and specific plane maintenance timelines.

Even though the model's 52.4% accuracy is low, these results are still incredibly useful. First, they provide a clear baseline proving that passenger demographics like age or gender have zero correlation with systemic airline delays, saving future analysts from wasting time on the wrong variables. Second, the feature importance metrics successfully pointed us toward what actually matters: geographical data like country names and airport codes.

References

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